

SECTION 2.8
SUBSPACES OF $\mathbb{R}^n$
ELEMENTARY LINEAR ALGEBRA

(MATH 2250-03)

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This section focuses on important sets of vectors in $\mathbb{R}^n$ called *subspaces*. Often, subspaces will arise in connection with some matrix A ; and they provide useful information about the equation $Ax = b$. The concepts and terminology in this section will be used repeatedly throughout the rest of the course.

Definition 1. A *subspace* of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that satisfies the following properties:

1. $0 \in H$ (i.e. H has an additive identity)
2. For each $u, v \in H$, we have $u + v \in H$ (i.e. H is closed under addition)
3. For each $u \in H$ and each scalar $c \in \mathbb{R}$, we have $cu \in H$ (i.e. H is closed under scalar multiplication)

Example 1. If v_1 and v_2 are in $\mathbb{R}^n$ and $H = \text{span}\{v_1, v_2\}$, then H is a subspace of $\mathbb{R}^n$. To verify this statement, note that the zero vector is in H (because $0v_1 + 0v_2$ is a linear combination of v_1 and v_2). Now take two arbitrary vectors in H , say,

$$u = s_1v_1 + s_2v_2 \quad \text{and} \quad v = t_1v_1 + t_2v_2.$$

Then, observe that

$$u + v = (s_1 + t_1)v_1 + (s_2 + t_2)v_2$$

which shows that $u + v$ is a linear combination of v_1 and v_2 , giving $u + v \in H$. Similarly, for any scalar $c \in \mathbb{R}$, we may see that

$$cu = cs_1v_1 + cs_2v_2$$

which is certainly a linear combination of v_1 and v_2 , giving $cu \in H$. So, H is a subspace!

In fact, given any set of vectors $v_1, \dots, v_p \in \mathbb{R}^n$, it may be shown (in a similar fashion to what was done in the previous example) that $\text{span}\{v_1, \dots, v_p\}$ is a subspace of $\mathbb{R}^n$. As such, we will refer to $\text{span}\{v_1, \dots, v_p\}$ as the *subspace spanned* (or *generated by*) $v_1, \dots, v_p$.

Also, one may note that $\mathbb{R}^n$ is a subspace of itself, as it satisfies the three properties required for a subspace. Similarly, the set consisting of only the zero vector in $\mathbb{R}^n$ is also a subspace of $\mathbb{R}^n$. Often, you may hear these subspaces referred to as the *trivial subspaces*, for obvious reasons.

COLUMN SPACE AND NULL SPACE OF A MATRIX

Subspaces of $\mathbb{R}^n$ usually occur in applications and theory in one of two ways, which we will now explore. In both cases, the subspace can be related to a matrix.

The first manner by which subspaces of $\mathbb{R}^n$ most commonly arise is by considering the *column space* of a matrix:

Definition 2. The *column space* of a matrix A , denoted $\text{Col}(A)$, is the subspace spanned by the columns of A . That is, if A is an $m \times n$ matrix with columns $a_1, \dots, a_n \in \mathbb{R}^m$, i.e. $A = \begin{bmatrix} a_1 & \dots & a_n \end{bmatrix}$, then

$$\text{Col}(A) = \text{span}\{a_1, \dots, a_n\}.$$

Example 2. Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$. Then, the column space of A is

$$\text{Col}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ -4 \\ -3 \end{bmatrix}, \begin{bmatrix} -3 \\ 6 \\ 7 \end{bmatrix}, \begin{bmatrix} -4 \\ -2 \\ 6 \end{bmatrix} \right\}.$$

Recall now that we have shown that the equation $Ax = b$ has a solution if and only if b is a linear combination of the columns of A . So, using the definition of the column space of A , this may be stated equivalently as " $Ax = b$ has a solution if and only if $b \in \text{Col}(A)$."

Considering that the column space arises in the context of a general inhomogeneous system $Ax = b$, it seems natural to consider the homogeneous system $Ax = 0$ as well. This line of thinking gives rise to the *null space* of a matrix:

Definition 3. The *null space* of a matrix A , denoted $\text{Nul}(A)$, is the set of all solutions to the homogeneous equation $Ax = 0$.

If A is an $m \times n$ matrix, it can be shown that $\text{Nul}(A)$ is, in fact, a subspace of $\mathbb{R}^n$. This is done by noting that $A0 = 0$ and by noting that, if $Av = 0$ and $Au = 0$, then $A(v + u) = Av + Au = 0 + 0 = 0$ and $A(cv) = cAv = c0 = 0$. More concisely, we have the following theorem:

Theorem 1. The null space $\text{Nul}(A)$ of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.

Broadly, if A is an $m \times n$ matrix, then by defining the linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $T(x) = Ax$, it follows that $\text{Col}(A)$ is a subspace of the range of T (i.e. $\mathbb{R}^m$), while $\text{Nul}(A)$ is a subspace of the domain of T (i.e. $\mathbb{R}^n$).

To test whether a given vector v is in $\text{Nul}(A)$, just compute Av to see whether $Av = 0$. Because $\text{Nul}(A)$ is described by a condition that must be checked for each vector, we say that the null space is defined *implicitly*. In contrast, the column space is defined *explicitly*, because vectors in $\text{Col}(A)$ can be explicitly constructed (by linear combinations) from the columns of A . To create an explicit description of $\text{Nul}(A)$, solve the equation $Ax = 0$ and write the solution in parametric vector form.

BASIS FOR A SUBSPACE

Because a subspace typically contains an infinite number of vectors, some problems involving a subspace are handled best by working with a small finite set of vectors that span the subspace. The smaller the set, the better. It can be shown that the smallest possible spanning set must be linearly independent.

Definition 4. A *basis* for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .

Example 3. The columns of an invertible $n \times n$ matrix form a basis for all of $\mathbb{R}^n$ because they are linearly independent and span $\mathbb{R}^n$, by the Invertible Matrix Theorem. One such matrix is the $n \times n$ identity matrix. Its columns are denoted by $e_1, \dots, e_n$:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

The set $\{e_1, \dots, e_n\}$ is called the *standard basis* for $\mathbb{R}^n$.

Example 4. To find a basis for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

we first write the solutions of $Ax = 0$ in parametric vector form. By row-reducing the augmented matrix corresponding to the homogenous system $Ax = 0$, we get

$$\left[\begin{array}{cc|ccc} A & 0 \end{array} \right] \xrightarrow{\text{Row Reduce}} \left[\begin{array}{cc|ccccc} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

This corresponds to the system of equations

$$\begin{aligned} x_1 - 2x_2 - x_4 + 3x_5 &= 0 \\ x_3 + 2x_4 - 2x_5 &= 0 \\ 0 &= 0. \end{aligned}$$

So, the general solution is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} = x_2 u + x_4 v + x_5 w.$$

The above shows that $\text{Nul}(A)$ coincides with the set of all linear combinations of u , v , and w : That is, $\{u, v, w\}$ generates $\text{Nul}(A)$. In fact, this construction of u , v , and w automatically makes them linearly independent, as the equation above shows that $0 = x_2 u + x_4 v + x_5 w$ only if the weights x_2 , x_4 , and x_5 are all zero. So, we may conclude that $\{u, v, w\}$ forms a basis for $\text{Nul}(A)$. That is, $\{u, v, w\}$ is a linearly independent set in $\mathbb{R}^5$ and

$$\text{Nul}(A) = \text{span}\{u, v, w\}.$$

Finding a basis for the column space of a matrix is actually less work than finding a basis for the null space. To compute a basis for the column space of a matrix A , it is helpful to exploit the following theorem:

Theorem 2. The pivot columns of a matrix A form a basis for the column space of A .

Let's see how this works by doing a quick example:

Example 5. To find a basis for the column space of the matrix

$$A = \begin{bmatrix} 2 & 1 & 6 & 7 \\ 3 & 8 & 9 & 10 \\ -4 & -9 & 0 & -3 \end{bmatrix},$$

we may simply compute the reduced row echelon form of A , which is

$$\begin{bmatrix} 1 & 0 & 0 & 12/13 \\ 0 & 1 & 0 & -1/13 \\ 0 & 0 & 1 & 34/39 \end{bmatrix}.$$

So, using the previous theorem, it follows that the first, second, and third columns of A form a basis for the column space of A . That is, the set of vectors

$$\left\{ \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \\ -9 \end{bmatrix}, \begin{bmatrix} 6 \\ 9 \\ 0 \end{bmatrix} \right\}$$

is linearly independent in $\mathbb{R}^3$ and

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \\ -9 \end{bmatrix}, \begin{bmatrix} 6 \\ 9 \\ 0 \end{bmatrix} \right\}.$$